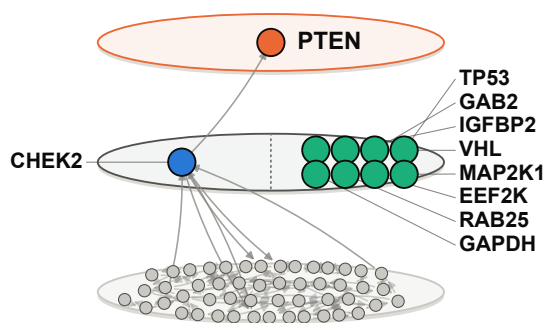
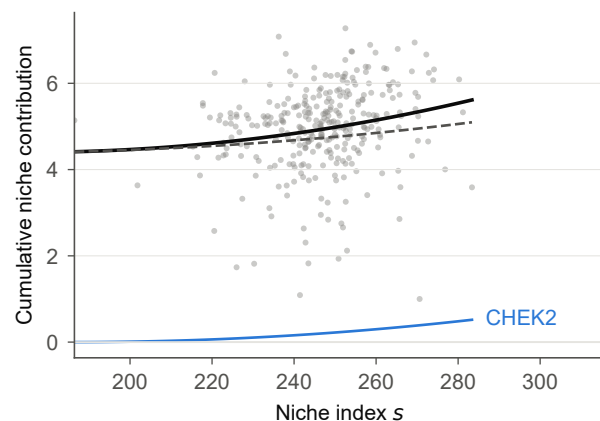


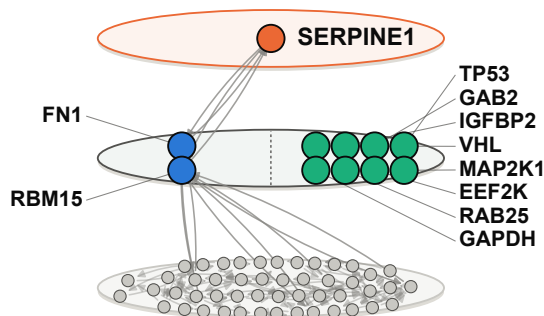
a PTEN joint / conditional rank 1 / 1



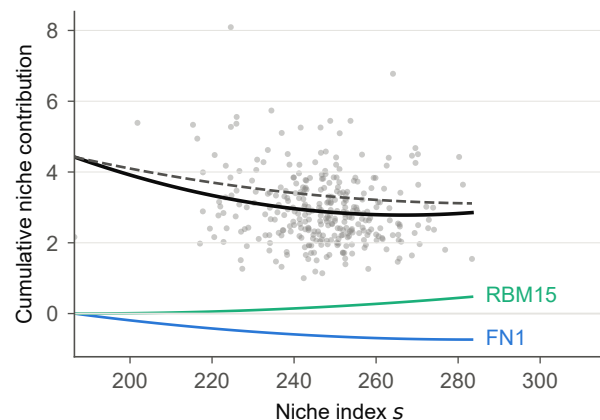
b PTEN curve decomposition



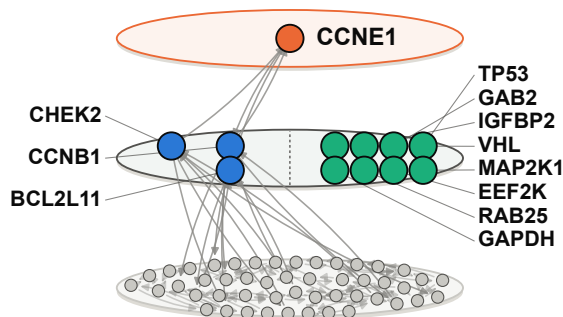
c SERPINE1 joint / conditional rank 2 / 2



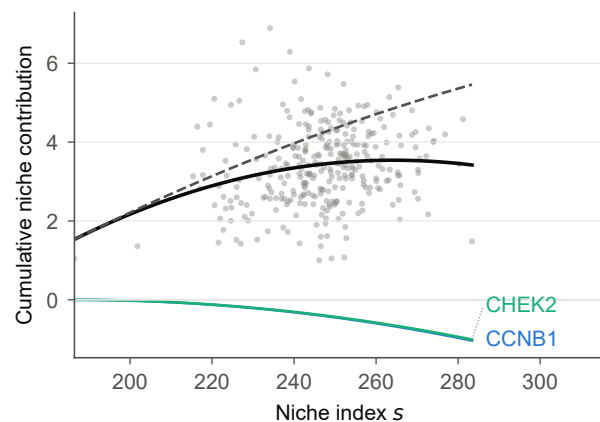
d SERPINE1 curve decomposition



e CCNE1 joint / conditional rank 2 / 2



f CCNE1 curve decomposition



● Exposure A — Directed LASSO support — Source contributions (labelled)
● Treatment proxy Z — Reconstructed curve ● Observed discovery protein
● Outcome proxy W - - - Intrinsic contribution